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Multiplicative Information Gating: Decomposing news into novelty, materiality, and polarity for calibrated selective forecasting

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Abstract: Converting a stream of textual events into a trading decision usually reduces each item to a single sentiment polarity and sums, conflating three properties that move prices through different channels: the direction of the news, how surprising it is, and how much it bears on value. We propose multiplicative information gating, in which each event is decomposed into polarity, novelty, and materiality and combined as their product, so that any near-zero factor vetoes the signal. A large language model supplies the three scores from calibration-anchored prompts; the resulting signal feeds a probabilistic forecaster whose output is calibrated and then passed through a conviction gate that may abstain. We prove that the product is the canonical combiner under per-axis multiplicative scaling, with veto and sign acting as consistency conditions, and we bound the signal by its weakest factor. The framework is market-agnostic; we instantiate it on exchange-traded equities using measured numbers only. On a live append-only ledger of 621 resolved forecasts, nominal 90% price intervals achieved 70.7% empirical coverage, confirming that deployed uncertainty is not self-correcting. An offline three-stage recalibration that first repaired a stacking domain mismatch cut expected calibration error from 0.35 to 0.05 without changing accuracy. Single-name next-period direction was unpredictable beyond drift (50.1% out-of-sample, ranking area under the curve near 0.47), so value had to come from selection rather than ordering: a conviction gate that abstains on roughly 94% of cases raised the realised 30-day hit rate to 60.6% against a 58.0% base (95% interval 58.8 to 62.4), winning in four of five test years. The contribution is a reusable, auditable recipe whose text and decision stages obey one selective principle.

Key words: Calibration, selective prediction, natural language processing, large language models, financial forecasting, sentiment analysis

1. Introduction

A growing class of decision systems reads text and acts on it: news headlines, filings, social posts, and support tickets arrive as timestamped streams, and somewhere downstream a model turns them into a number that drives an action. The dominant recipe scores each item for sentiment polarity, averages the scores, and feeds the average to a predictor [1, 2]; the financial literature has used it to show that media tone carries return-relevant information [1, 3]. But the recipe rests on an assumption that breaks down on the events that matter

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most.

Sentiment polarity answers one question: is this good or bad? It is silent on two others that a careful reader weighs instinctively: is the news actually new, or a restatement of something the market priced weeks ago, and does it bear on value, or is it cosmetic? A glowing but stale press release can score the same polarity as a quietly worded regulatory filing that threatens the firm's existence, even though their price impact differs by orders of magnitude. When such items are summed, the loud and empty ones drown out the rare informative ones, because averaging rewards volume over relevance.

We take the view that any textual event must possess at least three dissociable properties, and each influences price impact through an independent route: *polarity*, whether the underlying implied value goes up or down; *novelty*, whether the text imparts something genuinely new or whether we essentially already knew it; and *materiality*, how sensitive fundamental value is to the event (regardless of sign or novelty). Our approach, multiplicative information gating-preserves these three factors and multiplicatively integrates these values. Since the three values lie within bounded ranges (and at least two of them must be non-negative) and multiply together, the impact function works something like a logical AND. If the novelty and materiality of an item is zero, the product becomes zero regardless of its strength, so a single weak factor vetoes the whole event, and thus only fresh, meaningful information can yield a high-impact signal; in contrast, simply averaging the three values may yield a strong, high-impact signal from a stale or immaterial event that happens to possess a strong polarity.

Due to bound and non-negativity of at least two features, the product functions like a soft AND-gate: an stale or immaterial event will trigger near-zero signal even with strong polarity, so a single weak feature vetoes the entire event.

Simple averaging cannot achieve that, hence allows multiplicative signal to shield live position from noise just shouting its presence.

Two practical questions follow. First, who supplies the three scores? Hand-built dictionaries capture polarity passably [2] but struggle with novelty and materiality, which require world knowledge and context; we use a general-purpose instruction-tuned large language model, prompted with explicit numerical anchors so that its scores are reproducible and comparable across items [4, 5]. Second, once the gated signal enters a forecaster, how do we keep the system honest about what it does and does not know? The same gating principle reappears at the decision stage: a forecaster operating near the efficient-market limit [6] is mostly wrong about direction, so what it owes the user is not a confident guess on every name but a calibrated probability and the discipline to abstain when conviction is low [7–9]. Calibration and selective abstention are the decision-stage counterpart of the text-stage veto: each narrows what the system will commit to in return for being dependable on the rest.

We make the argument concrete on equities, where the data are public and the stakes are real, but the construction is market-agnostic and, we argue, domain-agnostic. The contributions are as follows.

1. We formalise multiplicative information gating: a decomposition of a textual event into polarity, novelty, and materiality, combined as a product, with a relevance-weighted aggregation to the entity level (Section 3). We prove a veto bound and show that, under five stated axioms whose substantive content is per-axis multiplicative scaling, the product is the canonical combiner (Section 3.2).

2. We give a reproducible scoring protocol in which a large language model returns the three axes from calibration-anchored prompts, with a deterministic lexicon fallback and persistent memoisation so that a given item is scored once (Section 3.3).

3. We couple the gated text signal to a calibrated, selective forecaster, and we report measured results only: live-ledger interval coverage, a three-stage calibration that repairs a stacking domain mismatch, a verified accuracy ceiling, and a conviction gate evaluated walk-forward over an eight-year span, 2018 to 2026 (Section 4).

Throughout, we are deliberate about what we can and cannot claim. The deployed ledger does not record per-event return labels, so we do not assert an isolated alpha for the text decomposition; its contribution is argued from its formal properties and its place in a pipeline whose end-to-end behaviour we do measure. Section 5 returns to this restraint.

2. Related work

Text and returns. The idea that media content carries return-relevant information is well established. Tetlock [1] linked pessimistic media tone to downward pressure and subsequent reversal; Antweiler and Frank [3] studied message-board activity; Loughran and McDonald [2] showed that finance-specific dictionaries matter because general-purpose ones mislabel domain terms. Supervised text models tailored to returns improve on dictionaries [10], and pre-trained language models, including finance-tuned variants, now dominate sentiment extraction [11, 12]. Most of this work, however, treats the per-item output as a scalar polarity and aggregates by counting or averaging; the novelty and materiality of an item are handled, if at all, by ad hoc filters rather than as first-class, separately scored quantities. Boudoukh et al. [13] is a notable exception in spirit: identifying which news is relevant sharply changes the measured information content. We make relevance and surprise explicit, continuous, and multiplicative. Where Ke et al. [10] learn return-predictive word weights by supervising directly on realised returns, we operate in the complementary regime in which no per-item return label is available at scoring time: the three axes are judged zero-shot, and the veto of Section 3.2 guarantees structurally that a stale or immaterial item cannot dominate, however the labels happened to fall.

Large language models as scorers. Instruction-tuned language models can follow rubric-style prompts and emit structured numeric judgements [4], and finance has adopted them quickly, from domain-specific pre-training at scale [14] to direct tests of whether a general model can read news and anticipate price moves [5]. The self-reported confidence of such models is itself imperfectly calibrated [15], which cautions against reading their raw outputs as probabilities. We do not ask the model to forecast; we ask it to *decompose*, a narrower and more checkable task, constrained by numerical anchors so that two runs on the same item agree. To our knowledge, no prior work scores novelty and materiality as separate, continuous, per-item quantities and makes them multiplicative preconditions for influence with a provable veto: existing pipelines, whether dictionary-based, supervised, or model-driven [2, 5, 10–12], reduce the item to one polarity before aggregation.

Calibration. Modern classifiers are often miscalibrated [16]; post-hoc remedies include Platt scaling [17] and isotonic regression [18], with reliability assessed by expected calibration error and the Brier score [19, 20]. Stacked generalisation [21] complicates this: a calibrator fitted on one stage's scores and applied to another silently breaks, a failure we diagnose and repair in Section 3.4.

Selective prediction. Allowing a model to abstain is an old idea [7] with a modern theory [8, 9] and a distribution-free cousin in conformal prediction [22–24]. The financial-machine-learning literature, by contrast, is dominated by accuracy and area-under-the-curve reporting, which a long methodological line has shown to be fragile under backtest overfitting and multiple testing [25, 26]. Deep and ensemble models for markets are now standard [27, 28], but few report whether their probabilities are trustworthy or whether they know when to stay out; our decision stage is built around exactly those two questions.

3. Materials and methods

3.1. The multiplicative information model

Let e denote a textual event (here, a news headline) about an entity (here, a listed equity). We assign e three scores: $s(e) \in [-1, 1]$ (polarity), $\nu(e) \in [0, 1]$ (novelty), and $\mu(e) \in [0, 1]$ (materiality). Polarity is the signed direction of the implied value change; novelty is the degree of genuinely new information, understood as surprise relative to the market's prior; materiality is the sensitivity of fundamental value to the event, regardless of sign. The *event signal* is their product,

$$a(e) \;=\; s(e)\,\nu(e)\,\mu(e) \;\in [-1, 1]. \tag{1}$$

We define the *relevance* of an event as $r(e) = \nu(e)\,\mu(e) \in [0, 1]$, so that $a(e) = s(e)\,r(e)$: the event signal is its polarity scaled by how novel-and-material it is. To aggregate the events $\mathcal{E}$ touching one entity into a single signal, we use a relevance-weighted mean restricted to events that clear a materiality floor μ_0,

$$A \;=\; \frac{\sum_{e \in \mathcal{E}:\, \mu(e) \geq \mu_0} r(e)\,a(e)}{\sum_{e \in \mathcal{E}:\, \mu(e) \geq \mu_0} r(e)} \;=\; \frac{\sum r(e)^2\, s(e)}{\sum r(e)}. \tag{2}$$

Relevance enters twice, once inside the event signal and once as the aggregation weight, so an item that is both novel and material dominates quadratically while cosmetic chatter is suppressed. The floor μ_0 discards items below a minimum materiality before weighting; we use $\mu_0 = 0.15$.

The motivation is a first-order account of price response: if novelty proxies the magnitude of surprise relative to the market's prior and materiality the sensitivity of value to the underlying variable, then to first order the expected value move is the product of a sign, a sensitivity, and a surprise magnitude. Additive sentiment keeps only the sign and discards the coupling, which is why it over-reacts to loud, stale, or immaterial items; (1) restores the coupling in its simplest form.

3.2. Why the product, and what it guarantees

The multiplicative form is not arbitrary. It is the simplest combiner consistent with how a careful analyst treats the three axes.

Proposition 1 (Veto bound) *For every event e, $|a(e)| \leq \min\{\, |s(e)|,\ \nu(e),\ \mu(e)\,\}$. In particular $a(e) = 0$ whenever any single factor is zero, and the aggregate satisfies $|A| \leq \max_e |s(e)|$.*

Proof Each factor lies in $[-1, 1]$ with $\nu, \mu \geq 0$, so $|a(e)| = |s(e)|\,\nu(e)\,\mu(e)$ is a product of three numbers of modulus at most one; such a product cannot exceed the modulus of any one of them, which gives the per-event bound and the zero case. For the aggregate, (2) gives $A = \big(\sum_e r(e)^2 s(e)\big) / \big(\sum_e r(e)\big)$, so

$$|A| \;\leq\; \frac{\sum_e r(e)^2\,|s(e)|}{\sum_e r(e)} \;\leq\; \max_e |s(e)| \,\cdot\, \frac{\sum_e r(e)^2}{\sum_e r(e)} \;\leq\; \max_e |s(e)|,$$

where the last step uses $\sum_e r(e)^2 \leq \big(\max_e r(e)\big) \sum_e r(e) \leq \sum_e r(e)$ because every $r(e) \in [0, 1]$. The weights $r(e)^2 / \sum_j r(j)$ are non-negative but sub-stochastic, summing to $\sum r^2 / \sum r \leq 1$; the aggregate is therefore a contraction toward zero of the polarities, never an amplification. $\square$

Proposition 1 is the formal content of "a single weak factor vetoes the event": a stale $(\nu \to 0)$ or immaterial $(\mu \to 0)$ item is capped near zero however extreme its polarity. No additive or convex combination of the three axes has this property, because a weighted sum $\alpha s + \beta\nu + \gamma\mu$ is generally nonzero when one term vanishes. We state the design choice as a short characterisation.

Proposition 2 (Characterisation) *Consider combiners $f(s, \nu, \mu)$ with $s \in [-1, 1]$, $\nu, \mu \in [0, 1]$ satisfying: (A0)* normalisation*, $f(1, 1, 1) = 1$; (A1)* sign fidelity*, $\operatorname{sign} f = \operatorname{sign} s$; (A2)* veto*, $f = 0$ if any of $|s|, \nu, \mu$ is 0; (A3)* separable monotonicity*, $|f|$ is non-decreasing in each of ν and μ for fixed others, and f is non-decreasing in s for fixed $\nu, \mu \geq 0$; and (A4)* factor homogeneity*, scaling any single argument by $\lambda \in [0, 1]$ scales f by λ. Then $f(s, \nu, \mu) = s\,\nu\,\mu$.*

Proof [Proof sketch] By (A4) applied to each argument in turn, $f(\lambda_1 s, \lambda_2 \nu, \lambda_3 \mu) = \lambda_1 \lambda_2 \lambda_3 f(s, \nu, \mu)$ for $\lambda_i \in [0, 1]$. Writing $s = \operatorname{sign}(s)\,|s|$ and taking $\lambda_1 = |s|, \lambda_2 = \nu, \lambda_3 = \mu$ relative to the unit corner gives $f(s, \nu, \mu) = |s|\,\nu\,\mu\, f(\operatorname{sign}(s), 1, 1)$. Sign fidelity (A1) forces $f(\pm 1, 1, 1) = \pm c$ with $c = f(1, 1, 1)$, and the normalisation (A0) fixes $c = 1$. Hence $f = s\,\nu\,\mu$; (A2) and (A3) are then automatically satisfied, so they are consistency requirements rather than the active constraints. □

The force of the result lies in homogeneity (A4): it requires each axis to act on its own multiplicative scale, and once that and the normalisation (A0) are granted, the product follows in two lines. The veto and monotonicity conditions (A2, A3) do not by themselves single out the product—many functions satisfy them—so we do not claim the product is the *only* sensible combiner, just the canonical one once per-axis scaling is imposed; and that scaling is exactly the property a position-taking system needs in order to treat freshness and relevance as preconditions for influence rather than as additive contributors.

3.3. Scoring the three axes with a language model

The decomposition is only as good as the three scores, which we obtain from a general-purpose instruction-tuned large language model (in our deployment, a compact production model, `Claude Haiku 4.5`), prompted as a quantitative analyst and constrained to return a JSON array of $\{\nu, \mu, s\}$ triples, one per headline, in input order. Two design choices make the scores reproducible.

Firstly, it pegs quantitative anchors to each of the three axes and forces the model to convert its subjective opinions into common metrics: for novelty, 0 corresponds to rehashes of market consensus already baked in for weeks, 0.2 to a result merely meeting consensus, 0.6 to an unexpected but modest development, and 1 to a sudden regulatory ban or fraud disclosure; for materiality, 0 corresponds to mundane boilerplate, 0.4 to a contract worth a few percent of revenue, 0.8 to a transformative deal, and 1 to an existential event; for polarity, -1 corresponds to extremely negative events such as fraud reports and covenant defaults, 0 to a neutral reshuffle, and $+1$ to events such as big earnings beats combined with raised guidance. Anchoring narrows the model output distribution and increases the chance of different model runs agreeing on the same event-a key property for this event definition, because the formula in (1) multiplies any disagreement on one axis by disagreements on other axes.

Secondly, each reported event is cached in a content-addressed database using the hash of the entity name and normalized headline: we don't recompute reports and therefore our runs are deterministic, with predictable cost. The system can also fall back to a dictionary lookup of pre-defined bullish and bearish wordlists to assign polarity, in which case we add neutral estimates to novelty and materiality (flagged for the consumer) to fall back when the model is down. The system architecture makes it trivial to swap any model: anchored on to a different model simply via instruction set.

3.4. From gated signal to calibrated probability

The aggregate A of (2) is one feature among several entering a hybrid forecaster that blends a gradient-boosted tree ensemble with a recurrent sequence model over price, momentum, trend, reversal, and volatility features to produce, for each name and horizon, a point forecast, a price interval, and a directional probability $\hat{p}$. We treat the forecaster as a black box; the object of study is not its architecture but its calibration and its willingness to abstain. A probability is only useful if it is calibrated: among names assigned $\hat{p} \approx 0.7$, about 70% should rise. We measure calibration by the expected calibration error over B equal-width bins,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|G_b|}{N} \left| \, \overline{y}_b - \overline{p}_b \, \right|, \tag{3}$$

where $\overline{p}_b$ and $\overline{y}_b$ are the mean predicted probability and mean realised outcome in bin b, and by the Brier score $\frac{1}{N}\sum_i (\hat{p}_i - y_i)^2$ [20].

Our first deployed model reported a next-day up-probability near 0.80 while its realised accuracy sat near 0.51, an ECE of 0.345 and a Brier score worse than a constant 0.5 forecast. The cause was a stacking domain mismatch [21]: an isotonic calibrator had been fitted on the out-of-fold *average of the base learners* but then applied to the *stacked* prediction, a different distribution, and the final blend was never itself calibrated. The repair is a three-stage calibration: (i) refit the isotonic map on the same base-average it was meant to correct, (ii) form the blend, and (iii) fit a final Platt scaler [17] on a held-out fold and apply it to both the test rows and the live row. Equation (3) is the diagnostic at each stage; the effect is reported in Section 4.2.

3.5. The conviction gate: abstention as decision-stage gating

Calibration makes $\hat{p}$ honest; it does not make it skilful. If short-horizon single-name direction is near-random (Section 4.3), a calibrated probability will sit near the base rate almost everywhere, and acting on every name is pointless. The remedy mirrors the text-stage veto: act only when conviction clears a threshold, and abstain otherwise. For a horizon of H trading days we emit an UP call when the calibrated horizon probability satisfies $\hat{p} \geq \tau^\star$, and NEUTRAL otherwise. The threshold is chosen on training folds as the smallest τ whose fired bucket reaches a target precision (here 0.60) while still firing at least a minimum fraction of the time (here 5%):

$$\tau^\star = \min \Big\{ \tau : \ \mathrm{prec}(\tau) \geq 0.60 \text{ and } \phi(\tau) \geq 0.05 \Big\}, \tag{4}$$

where $\mathrm{prec}(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$ is the fired-bucket precision and $\phi(\tau) = \Pr[\hat{p} \geq \tau]$ is the firing rate. (We reserve the word *coverage* for interval containment in Section 4.1; the firing rate ϕ is its selective-prediction analogue.) The gate is intentionally one-sided: confident DOWN calls on single names are destroyed by the upward drift of equities (Section 4.4), so the system emits long-or-neutral signals only. Like the event veto, the gate forgoes breadth for dependability, so the quantity worth reporting is precision at a given firing rate rather than accuracy or area under the curve.

Figure 1 summarises the two gates in one pipeline.

3.6. Data, deployment, and evaluation protocol

The system is deployed on liquid National Stock Exchange of India (NSE) large-capitalisation equities. Two bodies of evidence are used and kept strictly separate. The first is an *offline walk-forward* study on daily data

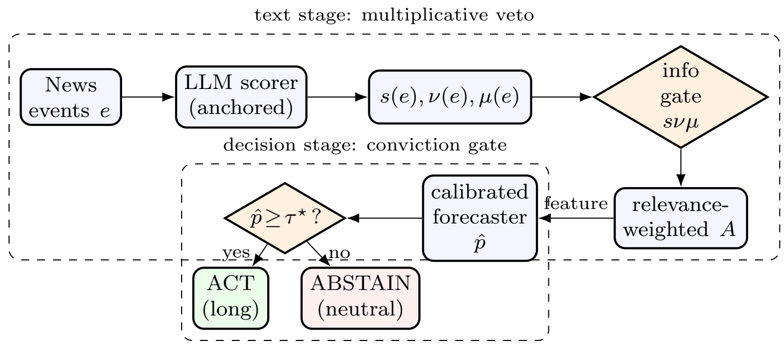

Figure 1. Two gates, one principle. The text stage decomposes each event and vetoes stale or immaterial items through the product $s\nu\mu$; the decision stage calibrates the forecaster's probability and abstains unless conviction clears $\tau^\star$; in both, breadth is given up for dependability. The schematic is conceptual.

from 2018 to 2026 for a 54-name universe, used to fit and evaluate the forecaster and the gate; all reported offline statistics are out-of-sample under an expanding window, so no future information enters a fitted fold. The second is a *live append-only ledger*: since April 2026 the deployed system has written every issued forecast (entity, timestamp, target date, point forecast, interval, nominal confidence, directional probability, horizon) to a database row that is later stamped with the realised price and outcome. The ledger is first-look only; a forecast is scored against the outcome recorded for its original target date and never retro-edited. It is the strongest evidence we have: produced by the running system, not reconstructed after the fact.

We report three kinds of measured result: interval coverage on the live ledger; the calibration repair and the accuracy ceiling from the offline study; and the selective gate's precision at its firing rate, also offline and walk-forward. No quantity below is illustrative unless its caption says so.

4. Results

4.1. Deployed uncertainty does not self-correct

Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE large caps (ABB, HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), each carrying a nominal 90% interval with a median half-width of 5.6% of price. Had the intervals been well calibrated, about 90% would contain the realised price; empirical coverage was 70.7% overall, a 19.3 percentage-point shortfall. The bulk of the forecasts were at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts; the smaller twenty-day and five-day buckets covered 86.8% and 100% (Table 1, Figure 2a). A deployed model's stated confidence was systematically overstated, and nothing in the pipeline noticed on its own: the empirical reason calibration and abstention cannot be afterthoughts.

Table 1. Live-ledger interval coverage by horizon (NSE large caps, 19 April–12 June 2026). All intervals carry a nominal 90% level.

Horizon	Resolved forecasts	Nominal coverage	Empirical coverage
5 trading days	8	90.0%	100.0%
10 trading days	575	90.0%	69.2%
20 trading days	38	90.0%	86.8%
All	621	90.0%	70.7%

4.2. A stacking domain mismatch, and a three-stage repair

The offline diagnosis of overconfidence is even more stark – and like the next two results, this is a walk-forward, not a live-ledger, result. The initial next-day directional head recorded probabilities near 0.80 vs a realized accuracy around 0.51 - an ECE of 0.345 and Brier score above 0.366, worse than guessing 0.5 consistently. Following the pipeline revealed the root cause detailed in Section 3.4: an isotonic map trained on the average of the base-learners was applied to the stack, but the final blend was entirely uncalibrated itself.After three fixes including rerunning the isotonic map in its intended space and adding a final Platt scaler on a held-out fold, the ECE dropped to 0.049 - bringing the reported up-probability back to reasonable range around 0.50 – while accuracy was unchanged: calibration changed what the model said about its confidence, not what it predicted.The takeaway for any stacking models is clear: any calibrator is tethered to its data distribution, and stacking is by definition mixing more than one distribution.

4.3. Single-name direction is unpredictable beyond drift

This "uncomfortable" reality, previously masked by prettier numbers, revealed itself under an honest calibration. By forcing probabilities back toward the base rate, we then directly posed: Does a single-name directional edge persist over the short horizon? A cross-sectional model designed to forecast direction for all names over a 5-day horizon achieved out-of-fold accuracy of 50.09%, with ranking areas-under-the-curve of around 0.49. In other words, this is chance. Five different methods for assessing this question, varying learning models, data inputs and market and trend regimes, yielded the same outcome; no stable edge could be sustained above the always-up base rate over daily time horizons. This outcome is consistent with the efficient-market prediction for liquid names [6], and it reframes the problem: because ordering names by predicted return is near-random, ranking-based value is unavailable, shifting the issue to one of selection, i.e. picking the few circumstances under which the probability is sufficiently deviating from the base rate to be exploitable.

4.4. Selection recovers a small, stable edge

Extending the horizon out to thirty trading days introduces two small edges: a positive equity trend (a positive thirty-day return about 58% of the time) and small idiosyncratic factor exposures extracted by a fitted cross-sectional model to factors that include momentum, trend, reversal, and volatility. If we walk forward the system from 2018 through 2026 on the 54-name universe ($91,471$ training observations and $50,272$ out-of-sample observations) the (4) threshold calibrates to $\tau^\star = 0.63$. The trained model selects the top 5.6% of observations for the UP call bucket and registers 60.6% of these as up, an edge over the base rate of 58.0% in directional accuracy of 2.6 percentage points, while abstaining the remaining 94% of the time.The fired UP calls sum to $2,840$ selections so the 60.6% hit rate carries a 95% binomial interval of $(58.8,\ 62.4\%)$ – a one-sided test against the 58.0% baseline gives $z \approx 2.9$ and $p \approx 0.002$, thus the effective edge is statistically significant at typical levels.That said, it is certainly not robust in a conventional sense – year-to-year the count of fired observations, and so the observed edge, can swing wildly between $+10.8$pp in 2025 and -2.4pp in 2024 (Table 2); we should interpret this as a small but reliably earned edge at the cost of rare occurrences. As one would expect, the area under the receiver operator curve for ranking is about 0.47 (a curve below the y=x line), indicating that the edge lives in the high-conviction tail, not in the ordering, which is what precision at a fixed firing rate registers and area-under-the-curve overlooks. Indeed the confidence gate returns more than a correct forecast on four of the five out-of-sample test years (Table 2, Figure 2b), the only exception being 2024. Confident DOWN calls by contrast hit around 10% of the time, undermined by the equity trend, which is why the gate never emits them.

Table 2. Walk-forward selective signal, by test year. Precision is the realised up-rate of the fired high-conviction bucket; base is the always-up rate; fires is the number of high-conviction calls.

Test year	Fires	Fired-bucket precision	Always-up base	Edge (pp)
2022	365	61.6%	54.0%	+7.6
2023	290	68.3%	65.3%	+3.0
2024	932	53.3%	55.7%	−2.4
2025	822	69.0%	58.2%	+10.8
2026	431	54.5%	47.1%	+7.4
Pooled	2,840	60.6%	58.0%	+2.6

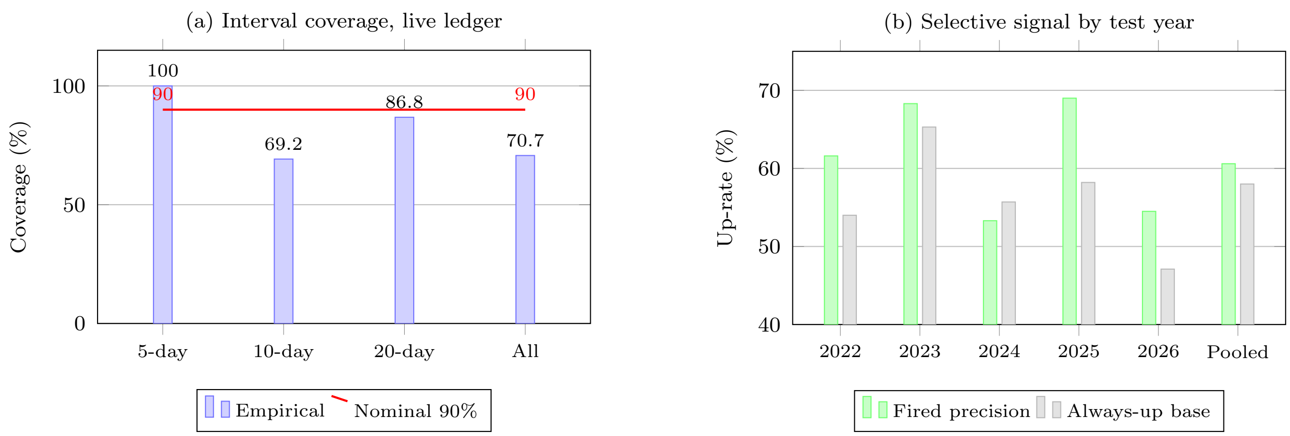

Figure 2. Measured behaviour of the two dependability mechanisms. (a) Empirical coverage of nominal 90% intervals on the live ledger by horizon (resolved forecasts: 8, 575, 38, 621 in total). (b) Fired-bucket precision of the conviction gate against the always-up base rate by test year, at about a 5.6% firing rate.

4.5. The veto in action

The text stage cannot be evaluated as an isolated alpha, because the deployed ledger does not store per-event return labels; Section 5 weighs this limit. What we can show is that the decomposition behaves as designed. Table 3 decomposes a small illustrative set of headlines: a glowing but routine in-line result and an emphatic but cosmetic announcement are both gated down, while a quieter disclosure that is genuinely new and material dominates despite its more moderate polarity—the protection Proposition 1 guarantees and additive sentiment cannot offer. The example is illustrative of the scoring behaviour and carries no performance claim.

5. Discussion

Every result here turns on the same trade-off. The text-stage veto refuses to let stale or immaterial items contribute, and the decision-stage gate stays out on the great majority of names: the system surrenders any pretence of commenting on everything and earns, in exchange, the right to be believed on the little it does flag. We think this is the correct posture for any model whose outputs are read literally by people with money or safety at stake.

Table 3. Illustrative decomposition of four headlines for one entity, following the anchors of Section 3.3; not a performance measurement.

Headline (paraphrased)	Polarity s	Novelty ν	Material. μ	Signal a
Quarterly profit in line with consensus, upbeat tone	+0.30	0.20	0.30	+0.018
Company reiterates previously announced buyback	+0.55	0.10	0.40	+0.022
Unexpected regulator notice on a core product line	−0.55	0.75	0.80	−0.330
Routine management reshuffle, no strategic change	+0.05	0.40	0.15	+0.003

Why multiplicative rather than additive. The veto bound of Proposition 1 is not a technicality; it is the whole point. Additive sentiment treats a loud restatement and a quiet revelation as comparable because it scores only tone. The product makes relevance a precondition for influence: we cannot attribute a standalone return to this choice on the present evidence, but the design consequence is testable wherever per-event labels exist—under the product, the events that move the aggregate are those an analyst would flag as new and consequential, and no others.

Generality. There is nothing about the method described in Section 3 that is peculiar to equities. For example, consider any series of timestamped textual events influencing a decision; these will have three same axes of a direction of implied effect, a degree of surprise, and a degree of impact. Triage of alerts in health care, priority management of security or ops tickets, and a moderation queue are susceptible to the same problem if they boil down to an overall urgency or sentiment score. They too could use the vetoing power over noise without substance, or noise that has ceased to be of immediate importance. We are not only able to apply this discipline at the decision level-calibrating before abstaining-but this discipline may also be carried to an arbitrary confidence threshold-set to deliver desired precision at a given reporting frequency. We have concentrated this research effort on a financial market-which is unforgiving in its results and readily accessible-but have phrased the present study as an example of applying these methods independently of specific features of the chosen venue.

Threats to validity and honest limitations. Four limits should be noted: First, the language model scores are judgements, not measurements – and although the tight binding by anchoring is to some degree helpful, we do not know to what degree anchoring helps make novelty and materiality objective-this scale is subject to change if, e.g. a different language model were used, or the prompt could drift. We believe that we limit some variance by using both a deterministic fallback and memoization. Second, the ledger we observe is very new-about two months of information, two months with seven firms; hence the coverage deficit, which we are convinced is real and significant at the important timescale, is an instance of the observed pipeline but not an eternal law. Third, and most importantly, because we do not have per-event return labels in deployment, we argue for the additional value of the decomposition based on the abstract properties and positioning in the observed pipeline. We are unable to show the effect in an isolation test, a difficult maneuver for which the critique of backtest overfitting [25, 26] would be quite applicable. Fourth, the number of cases of observed directional calls for which the outcome is known in the live ledger is only 28 so our live claim is built on the 621-forecast interval-coverage evidence as opposed to a live reliability diagram.

Relation to selective and conformal prediction. The decision gate is an instance of selective classification [7, 9]; what is new is pairing it with a probability whose calibration was itself repaired, and reading both stages as one gating principle. Distribution-free interval methods [22–24], adaptive conformal inference in particular, are built for exactly the non-stationary coverage failure of Section 4.1; integrating them is future work.

6. Conclusion

Turning text into decisions should not begin by collapsing every event to a single sentiment number. A textual event carries direction, surprise, and consequence, and these need not move together; taking their product instead of their sum builds a signal in which the weakest axis caps the rest, the safeguard that keeps a live decision from chasing loud noise. We proved that the product is the canonical combiner once per-axis multiplicative scaling is imposed, and bounded the aggregate by its strongest polarity. The same gating logic belongs at the decision stage: on a live deployment ledger, nominal 90% intervals covered only 70.7% of outcomes, so a forecaster has to be calibrated and willing to abstain. An offline three-stage recalibration that first repaired a stacking domain mismatch cut expected calibration error from 0.35 to 0.05 without touching accuracy; the corrected probabilities then exposed that short-horizon single-name direction is unpredictable beyond drift; and a conviction gate that stays out on about 94% of names raised the thirty-day hit rate to 60.6% against a 58.0% base, in four of five test years. What ties the two stages together is one rule: give up breadth where the signal cannot be trusted, and commit only where it can. Per-event labelled evaluation of the decomposition and adaptive conformal intervals for the price forecasts are the clear next steps.

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Declaration of generative AI and AI-assisted technologies

This study uses a large language model as a measurement instrument: an instruction-tuned model (Claude Haiku 4.5, Anthropic) scores news headlines for novelty, materiality, and polarity from the anchored prompt of Section 3.3; this use is part of the system under study, and all such outputs were validated against the deterministic fallback and persisted for audit. Separately, during manuscript preparation the authors used a large language model solely to improve the language and readability of author-written drafts; no generative tool produced data, results, figures, or interpretations. The authors reviewed and edited all content and take full responsibility for the publication.

Data availability statement

The live forecast ledger and the offline walk-forward statistics underlying Tables 1–2 and Figure 2 can be made available by the corresponding author on reasonable request, subject to the platform's terms. Source price data are from public NSE end-of-day records.

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